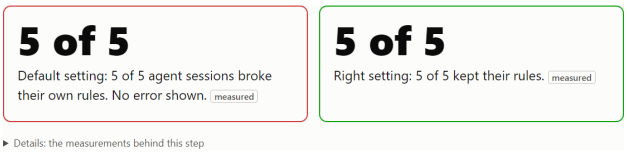
Silent context loss on local AI agents

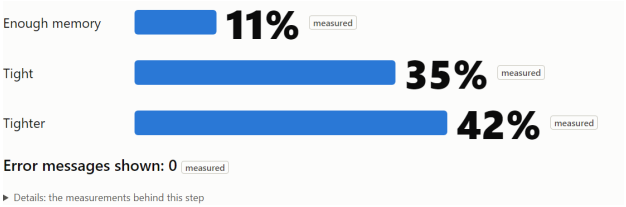
What we measured, what it costs, and the tool that picks the setting

Rithwik Sharma • DSE demo, 28 October 2026 • labels: measured run on our hardware priced, not executed token counts at list prices, no API call projected projected from another device or task size



1. Same laptop, two settings

- measured Default setting: 5 of 5 agent sessions broke their own rules. No error shown.[1]
- measured Right setting: 5 of 5 kept their rules.[2]



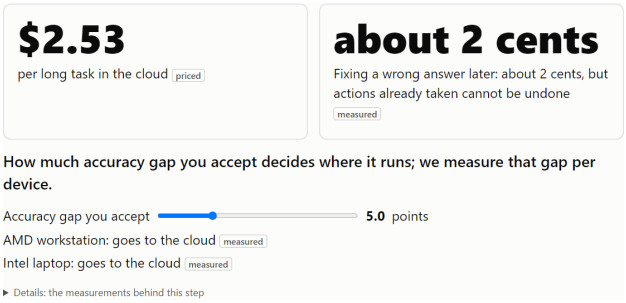
2. How often answers are silently wrong

- measured Enough memory: 11% wrong[3]
- measured Tight: 35% wrong[3]
- measured Tighter: 42% wrong[3]
- measured Error messages shown: 0[3]



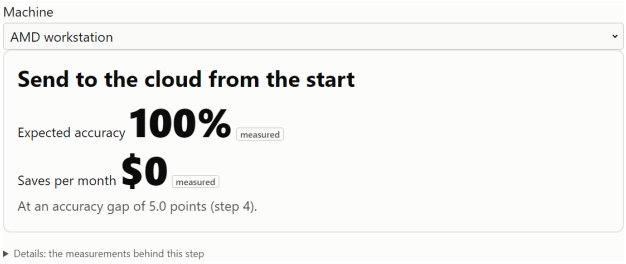
3. Why not just change one setting?

- measured Bigger memory setting: up to 16x slower, or will not start[4]
- measured The quick fix: still 30% wrong[3]
- measured Trimming old history: breaks a different rule[5]



4. What it costs

- priced per long task in the cloud[6]
- measured Fixing a wrong answer later: about 2 cents, but actions already taken cannot be undone[7]
- How much accuracy gap you accept decides where it runs; we measure that gap per device.



5. What our tool does

- measured AMD workstation: send to the cloud from the start at an accuracy gap of 5 points[8]
- measured Intel laptop: send to the cloud from the start at an accuracy gap of 5 points[8]
- not measured NVIDIA laptop

Still being measured.

- Local task accuracy on evo-t2s: the route decision uses the R2 session pass rate until the evo-t2s B5 task runs report task accuracy; the route-at-start decision covers qwen3:14b only.
- The escalation cost against claude-haiku-5-5: interim (its all-cloud price); the full-history redo ran on claude-sonnet-5-5 only.
- The evo-t2s Intel CPU trade-off is projected from evo-x2's 4096 tier, and the Blade is not measured; the evo-t2s setting comparison runs in its next access window.

Sources (row ids in docs/NUMBERS_REGISTER.md, or result file and line): 1 `t2s-r2-real-cpu`; 2 `t2s-r2-real-igpu`; 3 `x2-evalset-tasks-real-v2-wrong-rate`; 4 `impact-session-time-by-window`; 5 `R2-qwen3-14b-4096-reconciliation`; 6 `cloud-cost-per-real-task`; 7 `cloud-redo-full-context`; 8 `hybrid-route-at-start`.